

B*-tree floorplans

B*-trees are a popular compacted floorplan representation

The idea

- Root at the lower-left
- Walking left children grows the packing rightward
- Packing is typically done with a horizontal contour so y coordinates stay compacted
- Perturbing the tree, rotate, move, swap, yields neighboring packings for annealing later

Browser lab track

- Open **bstar-tree**
- Load a starter tree over A-E
- Step placement: parent, left child, right child
- Watch modules land left/right/above as the contour updates
- Inspect outline legality and deadspace
- Then code the same left/right packing rules

Implement track

- Build a B*-tree for the tiny modules
- Implement contour-based packing to emit (x, y, w, h)
- Verify non-overlap and outline containment
- Print the tree in a readable parenthesized form so goldens are reviewable
- Prefer deterministic child ordering for tests

Pitfalls

- Getting left/right geometry backwards is the number-one bug
- Contour updates that forget deleted segments leave modules floating or overlapping
- Don't confuse B*-tree adjacency with netlist adjacency

Your turn

- Produce a legal B^* -tree packing inside the outline
- Next: sequence pairs, another classic encoding with positive and negative sequences